

CHAPTER ONE

INTRODUCTION

Student identity is an important component of educational administration because institutions need reliable ways to recognize students and associate them with accurate institutional records. As institutions handle increasing amounts of student information, the process of identifying and verifying students becomes more than a matter of issuing physical identity cards. It involves maintaining accurate records, controlling access to identity information, and providing a practical way of confirming a student's identity whenever required.

The movement from manual record-keeping to computerized information systems has created opportunities to improve this process. Digital systems can organize student information in centralized databases, making records easier to retrieve, update, and manage. At the same time, developments in digital identity management provide a broader basis for understanding how an individual's identity can be represented and managed electronically. NIST's Digital Identity Guidelines, for example, discuss identity proofing, enrollment, authentication, and identity management as important components of digital identity services (Temoshok et al., 2025).

1.1 Background of the Study

Student identification is an important part of the administration of any educational institution because the institution must be able to establish who its students are and distinguish one student from another. In practice, identification is connected to activities such as admission, access to institutional services, academic administration, examination-related activities, and verification of student records. The traditional approach to student identification has generally depended on physical identity cards, registration numbers, photographs, and institutional records. Although these methods remain useful, their effectiveness depends heavily on the availability, accuracy, and condition of the physical and manually maintained records.

The increasing adoption of digital information systems has changed the way institutions manage information. Instead of relying entirely on physical records, institutions can maintain structured digital records that can be searched, updated, and retrieved electronically. Research on student information systems shows that system quality and information quality are

important to successful use and the management benefits obtained from such systems (Alharbi et al., 2022). In this context, the issue is no longer simply whether student information should be computerized, but how that information can be organized in a way that supports a specific institutional need effectively.

One of such needs is student identity verification. A student information system may contain a large amount of academic and administrative information, but that does not automatically make identity verification simple. An identification system must provide a practical connection between the person being identified and the corresponding institutional record. This makes digital identity management particularly relevant. Digital identity systems are concerned with establishing, maintaining, and using information that represents an individual in a digital environment. The National Institute of Standards and Technology explains digital identity in terms of processes such as identity proofing, enrollment, authentication, and identity management (Temoshok et al., 2025).

For an educational institution, this concept can be adapted to student identity management. A student's personal and academic attributes can be maintained as a structured digital identity, while an institutionally assigned registration number can provide a unique reference to the student record. Such an arrangement creates a more direct relationship between the student and the digital record held by the institution.

Another technology that has practical relevance to this problem is the Quick Response (QR) code. QR codes provide a machine-readable representation of information and are designed for rapid scanning and decoding. The current international specification, ISO/IEC 18004:2024, defines the characteristics, encoding methods, dimensional requirements, error correction rules, and decoding requirements of QR codes (International Organization for Standardization, 2024).

The use of QR codes in educational environments has already been explored in areas such as attendance management. Liew and Tan (2021), for example, developed a QR-code-based student attendance system and demonstrated how QR scanning could be used to automate the recording of student attendance. Their work also introduced additional checks involving classroom information, registered devices, and geolocation, showing that QR codes can serve as a practical bridge between a physical student activity and a digital institutional record.

A particularly relevant earlier study to the present project is the work of Kadri, Saheed, and Salau-Ibrahim (2016), who developed a QR-code-based student identity card for Al-Hikmah University. Their system linked student identification to a QR code and a server-based student record, allowing the student record to be accessed through QR scanning. Their work demonstrates the practicality of QR technology for student identification rather than limiting its use to attendance alone.

Despite the usefulness of physical student identification cards, they remain vulnerable to loss, damage, and outdated information. A physical card is also dependent on the ability of the person checking it to manually inspect the information displayed on the card. A centralized digital identity system offers a different approach because the displayed identity can be linked to information stored in an institutional database.

It is against this background that this project developed the **Digitization of Automated Student Identity System (DASIS)**. DASIS provides a centralized web-based environment for student registration and identity management. The system connects student information to a Digital Student ID and a QR code, allowing an authorized user to retrieve and verify a student's institutional identity electronically. Rather than treating registration, identification, and verification as disconnected activities, DASIS brings them together as parts of one identity-management process.

The completed system also provides administrative functions through which student accounts, Digital IDs, academic structures, verification activities, notifications, and audit records can be managed. In this way, the project moves beyond merely generating a QR code and instead presents a complete student identity-management workflow. The system therefore addresses the practical problem of organizing student identity information while providing a faster method of verifying that information.

1.2 Statement of the Problem

Student identification within educational institutions is often carried out using physical identification cards, registration numbers, and manual checking of institutional records. Although these methods can identify students, they become less convenient as the number of student records increases. An administrator or verification officer may need to inspect a

physical card or manually search through institutional records before confirming the identity of a student.

The problem becomes more significant when physical student identification cards are lost, damaged, or contain information that has changed. A change in department, programme, level, or other student information may require the identity information maintained by the institution to be updated. Where identification depends heavily on physical records, such changes may create inconsistencies between the information presented by a student and the information maintained by the institution.

Another problem is the separation between student records and the verification process. A student may have a valid record in an institutional database, yet the process of confirming that record may still require manual searching. This creates an unnecessary gap between storing student information and using that information for real-time verification.

Existing research shows that QR technology can reduce the manual effort involved in retrieving or recording student-related information. However, many educational applications of QR codes concentrate on attendance rather than identity management. Liew and Tan (2021), for example, focused on attendance automation, while Kadri et al. (2016) demonstrated the application of QR technology specifically to student identification.

The problem addressed by this study is therefore the absence of an integrated and convenient approach to student identity management in which student registration, centralized identity records, Digital Student IDs, QR-based verification, and verification records are managed as connected functions within one system.

1.3 Aim and Objectives of the Study

1.3.1 Aim

The aim of this project is to design and implement a **Digitization of Automated Student Identity System (DASIS)** that improves the management and verification of student identities through centralized digital records, Digital Student IDs, and QR-code technology.

1.3.2 Objectives

The specific objectives of the study are to:

1. Review relevant literature on student identity management systems, digital identification, QR code technology, and automated identity verification.
2. Design the proposed system using appropriate system architecture, Entity Relationship Diagram (ERD), and Data Flow Diagram (DFD) for managing and verifying student identities.
3. Implement the proposed system to register students, manage their identity records, generate digital student IDs with QR codes, and provide automated identity verification.
4. Test and evaluate the developed system to determine its functionality, usability, accuracy, and effectiveness in improving student identity management and verification.

1.4 Research Questions

The study seeks to answer the following questions:

1. What challenges are associated with the traditional management and verification of student identities?
2. How can digital identity and QR code technology improve student identity verification?
3. How can a web-based system integrate student registration, Digital Student IDs, and automated verification into a single platform?
4. How effective is DASIS in improving the management, retrieval, and verification of student identity information?

1.5 Significance of the Study

The significance of the study lies in the way DASIS connects student identity management with digital verification. Rather than viewing identification as simply the production of an identity card, the project treats it as a continuous process of creating, maintaining, retrieving, and verifying a student's institutional identity.

For the **educational institution**, the system provides a centralized environment for managing student identity information. Student records can be stored electronically and retrieved when required, reducing dependence on manual record searches.

For **students**, the system provides a Digital Student ID that can be accessed electronically and associated with a QR code. This gives students a convenient digital representation of their institutional identity.

For **administrative staff**, DASIS provides tools for registering students, managing account status, controlling Digital IDs, managing academic structures, and reviewing identity-related records.

For **identity verification officers**, the QR verification functionality provides a faster mechanism for retrieving the student record associated with a Digital ID. This addresses one of the main problems identified in the study: the time and effort required to manually confirm student information.

For **future researchers and developers**, the project provides a practical example of how digital identity concepts and QR technology can be applied to student identity management. The project can also serve as a starting point for future improvements such as biometric verification or mobile Digital ID applications.

1.6 Scope of the Study

The study covers the design and implementation of a web-based student identity management and verification system.

The scope includes student registration and management, storage of personal and academic information, Digital Student ID management, QR-code generation and scanning, identity verification, verification history, academic-structure management, notifications, and audit records.

The system provides separate functionality for students and administrators. Students can access their identity-related information and verification services, while administrators can manage student records and other identity-related functions.

The scope does not include academic result processing, course registration, examination management, attendance management, financial management, or biometric identification. This distinction is important because the purpose of DASIS is specifically student identity

management and verification rather than a complete student information management platform.

1.7 Organization of the Work

The project is organized into five chapters.

Chapter One introduces the study by discussing the background, problem, aim and objectives, research questions, significance, scope, organization of the work, and key terms.

Chapter Two reviews relevant literature and related studies on digital identity, student identity management, QR technology, and automated verification. It also establishes the theoretical basis of the study and identifies the gap addressed by DASIS.

Chapter Three explains the methodology adopted for the project and presents the design of DASIS, including its architecture, database, data design, and interface considerations.

Chapter Four presents the implementation of DASIS, the major system interfaces, implementation procedures, algorithms, testing activities, and evaluation results.

Chapter Five discusses the overall findings, presents the conclusion and recommendations, explains the contribution of the project, and identifies areas for future research.

1.8 Definition of Terms

1. **Automated Identity Verification:** The computerized process of confirming whether a presented identity corresponds to a valid identity record stored in a system.
2. **Digital Identity:** A collection of information associated with an individual and used to represent that individual within a digital environment.
3. **Digital Student ID:** An electronic representation of a student's institutional identity containing relevant identification information.
4. **QR Code:** A two-dimensional machine-readable code used to encode information that can be retrieved through an appropriate scanning device.
5. **Student Registration Number:** A unique institutional identifier assigned to a student.
6. **Student Identity Management:** The process of creating, storing, maintaining, updating, and verifying information used to identify students.

7. **Identity Verification:** The process of confirming that a person or presented identity corresponds with the expected identity information.
8. **Digitization:** The conversion of information or processes from physical or manual form into digital form.

CHAPTER TWO

LITERATURE REVIEW

This chapter examines existing knowledge and research relevant to the Digitization of Automated Student Identity System (DASIS). The review is based on the understanding that the development of a student identity system should build on existing work rather than operate independently of it. The chapter therefore discusses the concepts that provide the foundation for digital student identity management, examines the theories relevant to digital identity and verification, reviews previous studies that have applied digital systems and QR technology in educational environments, and identifies the gap that the present study addresses.

2.1 Conceptual Framework

The conceptual basis of DASIS begins with the idea that student identification is not simply about possessing a card. It involves the creation of an identity record, maintaining that record, and providing a reliable means of confirming the record when required.

2.1.1 Student Identity Management

Student identity management involves the collection and maintenance of information that distinguishes one student from another. Within an educational institution, this may include a student's name, registration number, photograph, faculty, department, programme, level, academic session, and other relevant personal information.

Traditional approaches generally distribute this information across physical identity cards, paper documents, and institutional records. The problem with such an arrangement is not necessarily that the information is unavailable, but that accessing and maintaining it can require multiple manual steps.

Digital student identity management changes this relationship by making the identity record a structured digital object. Information can be stored centrally, connected through database relationships, updated when required, and retrieved by authorized users.

This is consistent with the broader concept of digital identity management. NIST's Digital Identity Guidelines describe digital identity systems in terms of identity proofing, enrollment, authentication, and management processes (Temoshok et al., 2025). Although the NIST

framework is broader than student identification, its underlying principle is relevant: an identity must be established and managed in a controlled manner so that it can be relied upon when authentication or verification is required.

For DASIS, this concept is reflected in the relationship among the student account, the student record, and the Digital Student ID.

2.1.2 Digital Student Identification

Digital student identification refers to representing a student's institutional identity electronically rather than relying exclusively on a physical identification card.

The value of digital identification becomes clearer when the identity is connected directly to the institution's underlying database. Instead of treating the information printed on an identity card as the final source of truth, the digital system can retrieve current information from the centralized student record.

This distinction is important to DASIS. The Digital Student ID is not implemented merely as a visual replacement for a physical card. It is connected to the student's underlying identity record and can therefore participate in the verification process.

The conceptual advantage is that the identity representation and the institutional record remain connected. When the status of a student or Digital ID changes, the system can reflect that change during subsequent verification.

2.1.3 QR Code Technology

QR codes are two-dimensional machine-readable symbols designed to encode information in a form that can be captured and decoded by compatible devices. ISO/IEC 18004:2024 defines the technical characteristics, encoding methods, symbol formats, error-correction rules, decoding requirements, and quality requirements associated with QR codes.

The practical significance of QR technology is its ability to create a bridge between a physical object or displayed code and a digital information system. In an identity-management environment, a QR code can be associated with a particular identity record so that scanning the code provides a starting point for retrieving that record.

This approach has already been demonstrated in educational environments. Kadri et al. (2016) developed a QR-code-based student identity card in which student identification was associated with a server-based student record. The study showed that scanning the code could provide access to the student's academic information and create a more direct relationship between the physical identification medium and the digital record.

Similarly, research on QR-code-based attendance systems demonstrates the usefulness of QR codes for reducing manual processing in educational environments. Liew and Tan (2021) showed that QR codes could support automated student attendance while incorporating additional checks to strengthen the process.

These applications suggest that the value of QR technology is not limited to the information stored directly inside the code. Its greater value comes from using the code as a mechanism for connecting a physical interaction with a trusted digital record.

2.1.4 Automated Identity Verification

Identity verification involves determining whether the identity presented during an interaction corresponds to a valid record. In a computerized system, the process normally involves retrieving an identity record, evaluating the submitted identifier or evidence, and returning a result.

NIST describes identity proofing as a process involving resolution, validation, and verification of a claimed identity (Temoshok et al., 2025). While DASIS is not a national identity-proofing platform, the conceptual relationship is useful because the project also depends on associating submitted identity information with a known institutional record.

Within DASIS, the QR token or registration number functions as the starting point for this process. The system retrieves the associated student record, examines the relevant identity conditions, and produces a verification result.

The important conceptual distinction is therefore between **identification** and **verification**. Identification answers the question, “*Which student record does this identifier refer to?*” Verification goes further by determining whether the retrieved identity is valid under the conditions applied by the system.

2.1.5 Student Information Management Systems

Student information systems have become important components of modern educational administration because they allow institutions to manage substantial amounts of student information electronically. Research into student information systems has shown that system quality and information quality can influence system use, satisfaction, and resulting management benefits (Alharbi et al., 2022).

However, a general student information system and a dedicated student identity system are not exactly the same thing. A general information system may manage academic records, registration, results, courses, and other institutional activities, while identity verification may receive less attention.

This distinction provides an important conceptual foundation for DASIS. The project concentrates on one specific institutional problem—student identity management—and organizes its major functions around registration, Digital ID creation, verification, and identity-related administration.

2.2 Theoretical Framework

The theoretical framework of DASIS is based primarily on **Digital Identity Management** and **Identity Verification**.

2.2.1 Digital Identity Management Theory

The digital identity management concept explains how information representing an individual is created, maintained, associated with a digital account, and used within a digital system.

A useful way to understand the concept is to view identity as more than a collection of personal data. The information must be associated with an identifiable subject, maintained over time, and controlled so that it can be used appropriately. NIST's Digital Identity Guidelines emphasize identity proofing, enrollment, authentication, and identity management as interconnected parts of digital identity services (Temoshok et al., 2025).

The relevance of this theory to DASIS lies in the system's treatment of student identity as a managed digital record. A student account is connected to a student profile; the profile is connected to academic information; and the Digital ID is connected to the student's identity

record. The result is an identity structure that can be maintained rather than a static card that exists independently of the institution's database.

2.2.2 Identity Verification Theory

Identity verification focuses on the process of determining whether a claimed or presented identity corresponds to the expected identity. In the context of DASIS, the theory is represented operationally. A student identity is registered in the database, an identifier is associated with that identity, and a later verification request uses the identifier to retrieve the record and determine whether the identity can be confirmed. The QR code component, therefore, does not replace identity verification; it supports the retrieval process that makes verification faster and more convenient.

The theoretical relationship can be expressed as:

Student Identity Record → Identifier → Retrieval → Validation → Verification Result

This provides the conceptual basis for the QR verification workflow implemented in DASIS.

2.2.3 Relationship Between the Theories and DASIS

Digital Identity Management explains how the student's identity is created and maintained, while Identity Verification explains how that identity is confirmed when required. DASIS combines the two concepts. During registration, identity information is collected and stored as a structured digital identity. During verification, the identifier associated with that identity is used to retrieve the record and determine whether the identity remains valid.

This relationship provides the theoretical justification for the project's central design: the Digital Student ID and QR code are not independent features but components of a wider identity-management and verification process.

2.3 Review of Related Studies

2.3.1 Kadri, Saheed and Salau-Ibrahim (2016)

Kadri, Saheed, and Salau-Ibrahim developed a **student identity card based on advanced QR code technology** at Al-Hikmah University. The study addressed the limitations of conventional student identification by connecting a QR code to a server-based student record. When the QR code on the student identity card was scanned, the associated information could be retrieved from the database.

The importance of this study to DASIS is its direct focus on **student identification rather than attendance**. It demonstrates that QR codes can be used as a link between the physical presentation of a student's identity and a digital institutional record.

However, the earlier system concentrated primarily on the QR-enabled identification card. DASIS extends the concept into a broader management platform by integrating student registration, Digital ID management, QR verification, account management, verification history, notifications, and administrative controls.

2.3.2 Liew and Tan (2021)

Liew and Tan developed a **QR Code-Based Student Attendance System** for higher education. Their work responded to the difficulty and time involved in manual attendance recording and introduced a mobile QR-based mechanism for capturing attendance. The system included checks involving class information, the registered mobile device, and geolocation. The authors reported that the system addressed limitations associated with the existing attendance process.

The study demonstrates an important principle: QR codes can reduce manual interaction with institutional records. However, the problem being solved was attendance rather than student identity management.

This distinction is important for DASIS. The present project adopts the efficiency advantage demonstrated in QR-based educational systems but applies it specifically to the problem of identity management and verification.

2.3.3 Wenny, Suhartono, and Parenreng

Wenny, Suhartono, and Parenreng developed a QR-code-based lecture attendance system to address manual attendance collection, human error, and attendance fraud. Their research included requirements analysis, development, testing, evaluation, and use of the resulting system.

The study reinforces the argument that QR technology can replace repetitive manual processes in educational environments. At the same time, its focus remains attendance. The

present study differs by using QR technology as an identity-verification mechanism connected to a student's Digital ID.

2.3.4 Shaban et al.

Shaban and colleagues developed a QR-code-based student attendance application in a higher education setting. The application enabled students to scan a unique QR code during a class session and have their attendance recorded in the learning-management environment. The study demonstrates the practical acceptability of QR-based interactions among students and shows that such technology can provide immediate automated recording.

Again, the study supports the feasibility of QR technology in an educational environment, but its purpose differs from DASIS. DASIS is concerned with identifying students rather than simply recording participation in an event.

2.3.5 Student Information Systems in Nigerian Higher Education

Research on student information systems has also examined how such systems contribute to institutional management. Alharbi et al. (2022) investigated student information systems using an empirical model involving instructors from Nigerian universities. Their findings indicated that system quality and information quality influence system use and user satisfaction and contribute to management benefits.

This work is relevant because it emphasizes that simply computerizing records is not enough. The usefulness of the resulting system depends on the quality of the system and the information it provides.

DASIS reflects this principle by concentrating on structured identity records, controlled access, verification records, and mechanisms that allow administrators to maintain student identity information.

2.3.6 Digital Identity in Student Record Management

Research by Aalto University on students' digital identities in student record management examined the relationship between accounts, digital identity, personal identity, and identity management in a student-record environment. The study demonstrates that a student's digital identity should be understood as part of a broader information system rather than as an isolated username or account.

This perspective is particularly useful to DASIS because the project also treats identity as an interconnected set of information. The student account, personal details, academic structure, Digital ID, and verification history are related rather than independent.

2.4 Summary and Research Gap

The literature demonstrates that digital systems can improve the management of student and identity-related information. Research on student information systems shows the value of centralized digital records, while QR-code studies demonstrate the ability of QR technology to reduce manual processing in educational environments.

The review also shows that QR codes have already been applied directly to student identification. Kadri et al. (2016) demonstrated that a QR-enabled student identity card could connect a student with a server-based institutional record.

However, most of the reviewed QR-based educational applications focus on **attendance**, while systems dealing directly with student identification tend to focus on the identity card itself. The literature therefore reveals a practical gap: there is room for a system that treats the student's digital identity, QR-based verification, identity records, and administrative management as parts of one integrated platform.

DASIS addresses this gap by combining:

- centralized student identity records;
- Digital Student IDs;
- QR-code-based identity verification;
- registration-number verification;
- student account management;
- Digital ID status management;
- verification history;
- administrative notifications; and
- audit records.

The contribution of the present project is therefore not simply the use of QR codes. The contribution lies in **integrating QR-based verification into a broader student identity-management system.**

CHAPTER THREE

DESIGN METHODOLOGY

The MAPOLY guideline describes Chapter Three as the section where the methodology and technical design of the solution are presented, including architecture, data models, algorithms, interface design, and related diagrams.

3.1 Methodology

The **Waterfall Software Development Model** was adopted for the development of DASIS.

The choice of the Waterfall model was influenced by the relatively well-defined nature of the project requirements. The major functions of the system—student registration, identity management, Digital ID generation, QR verification, and administrative management—could be identified before implementation. This made it possible to organize development into clearly defined stages.

The first stage was **requirements analysis**. At this stage, the student identification problem was examined, and the requirements of the proposed solution were identified. The major requirements included student registration, identity record management, Digital ID creation, QR verification, and administrative control.

The second stage was **system design**. The results of the requirements analysis were translated into the system architecture, database structure, data flow, interface structure, and operational processes.

The third stage was **implementation**. The system was developed using HTML, CSS, JavaScript, PHP, and MySQL. The frontend was developed to provide the user interface; JavaScript was used for client-side interaction; PHP handled the server-side API operations; and MySQL provided persistent data storage. These are the technologies already identified in the original project methodology.

The fourth stage was **testing**. Individual modules and complete user workflows were tested to determine whether the implemented system behaved according to the defined requirements. Testing covered authentication, student registration, student management, Digital ID operations, QR verification, verification history, notifications, and administrative controls.

The final stage was **deployment and maintenance**. The completed system was configured to operate in the intended web environment. Maintenance considerations include correcting defects, updating system components, maintaining the database, and adding future functionality where necessary.

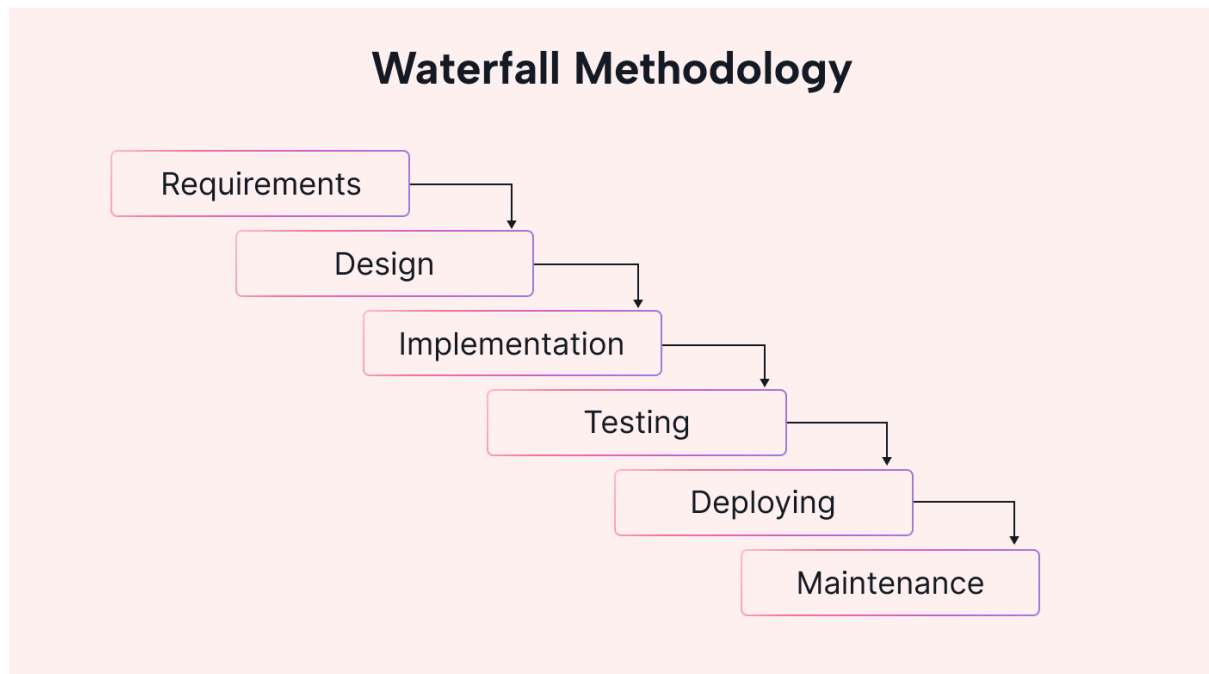


Figure 1: Waterfall software development model adopted for DASIS.

The adopted methodology provided a structured sequence from requirements through design, implementation, testing, and deployment. This was appropriate for the project because each major stage produced information required by the subsequent stage.

3.2 Choice of Development Tool (Programming Language)

The choice of development tools was based on the nature of the project and the requirements of a database-driven web application.

1. **HTML:** HTML was used to structure the DASIS interface. Forms, tables, navigation areas, buttons, student-information sections, and other interface elements were constructed using HTML.
2. **CSS:** CSS was used to control the visual presentation of the system. It was used for layout, spacing, typography, responsiveness, buttons, cards, status indicators, navigation, and other visual components.

3. **JavaScript:** JavaScript was used to provide interactive behaviour on the client side. It handles interface interactions, page rendering, form behaviour, communication with the backend API, notification display, and QR code scanning functionality.
4. **PHP:** PHP serves as the primary server-side programming language in DASIS. It handles user authentication, student registration, database communication, student management, Digital ID operations, QR verification, notifications, and administrative functions.
5. **MySQL:** MySQL was selected as the relational database management system because DASIS contains several related categories of data. The database stores user accounts, student records, faculties, departments, programmes, academic sessions, Digital IDs, verification logs, notifications, and audit records.
6. **Visual Studio Code:** Visual Studio Code was used as the development environment for writing, organizing, and maintaining the source code.
7. **XAMPP:** XAMPP provided the local development environment, including the Apache web server and MySQL database server, used during development and testing.

The technology selection is consistent with the approach described in the original methodology, which identified HTML, CSS, JavaScript, PHP, MySQL, Visual Studio Code, and XAMPP as the principal development tools.

3.3 System Design and Data Collection

3.3.1 System Design

DASIS was designed around the relationship between a student and a centralized institutional identity record.

The process begins when a student is registered. Personal and academic information is captured and stored in the database. The student record is associated with a user account and becomes the basis for the student's Digital Identity within the system.

A Digital Student ID is associated with the student record. The Digital ID contains identification information and a unique QR-related identifier.

During verification, an authorized user scans the QR code or provides a registration number. The system uses the supplied information to retrieve the corresponding student record and determines whether the identity is valid under the system's verification conditions.

The result of the verification is recorded so that the verification history can be reviewed later.

The design therefore follows the relationship:

Registration → Student Record → Digital ID → QR Identifier → Verification → Verification Record

This process is consistent with the system process established in the earlier methodology, but the present version reflects the actual implemented system rather than describing it as a future design.

3.3.2 Data Collection

Data requirements for DASIS were established through observation of student identification procedures, examination of the information required to identify students, and review of literature on digital identity management and QR-based verification.

The collected information was used to determine:

- The information required during student registration;
- The information required by a Digital Student ID;
- The relationships among students and academic structures;
- The data required for verification;
- The information required for administrative management; and
- The information required for verification and audit records.

The study therefore used both **direct observation of the identity-management problem** and **secondary information from existing literature and related systems**.

3.4 Design Architecture

DASIS adopts a three-tier architecture consisting of the **presentation layer, application layer, and database layer**. This separation enables the interface, processing logic, and stored data to perform distinct responsibilities.

1. Presentation Layer

The presentation layer is the part of the application with which students and administrators interact. It contains the login interface, student dashboard, administrator dashboard, forms, tables, Digital ID displays, QR verification interface, notification interface, and other visible system components. HTML, CSS, and JavaScript provide the main technologies used at this layer.

2. Application Layer

The application layer contains the server-side processing logic of DASIS. PHP handles authentication, student management, Digital ID operations, verification requests, notifications, academic structure management, and communication with the database. This layer also provides the API endpoints through which the frontend communicates with the backend.

3. Database Layer

The database layer uses MySQL to store the persistent data of the system. It contains the structured records required to operate DASIS, including users, students, academic structures, Digital IDs, verification records, notifications, and audit records.

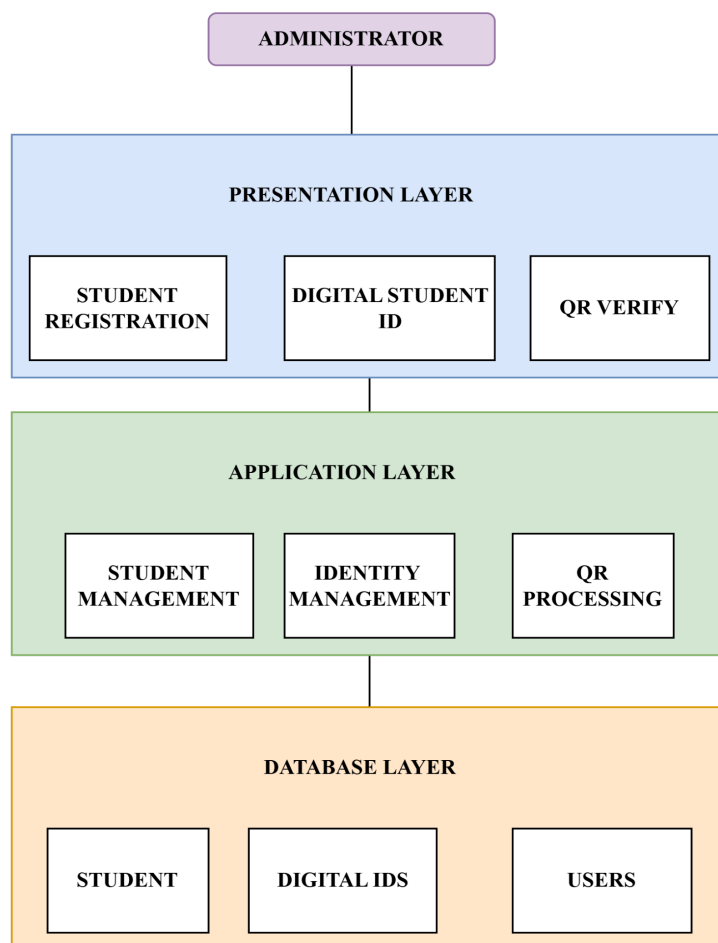


Figure 2: Three-tier architecture of the Digitization of Automated Student Identity System.

The architecture demonstrates the flow of information from the user interface through the application layer to the database and back to the user. This separation also makes it easier to maintain or modify individual parts of the application.

3.4.1 Database Design

The database design uses a relational structure in which related information is stored in separate tables and connected through keys.

The actual DASIS database is organized around the relationship between users and students. A student account is represented in the **users** table while the student's institutional details are represented in the **students** table.

The academic structure is further divided into faculties, departments, programmes, and academic sessions. This prevents repeated storage of the same academic information across multiple student records.

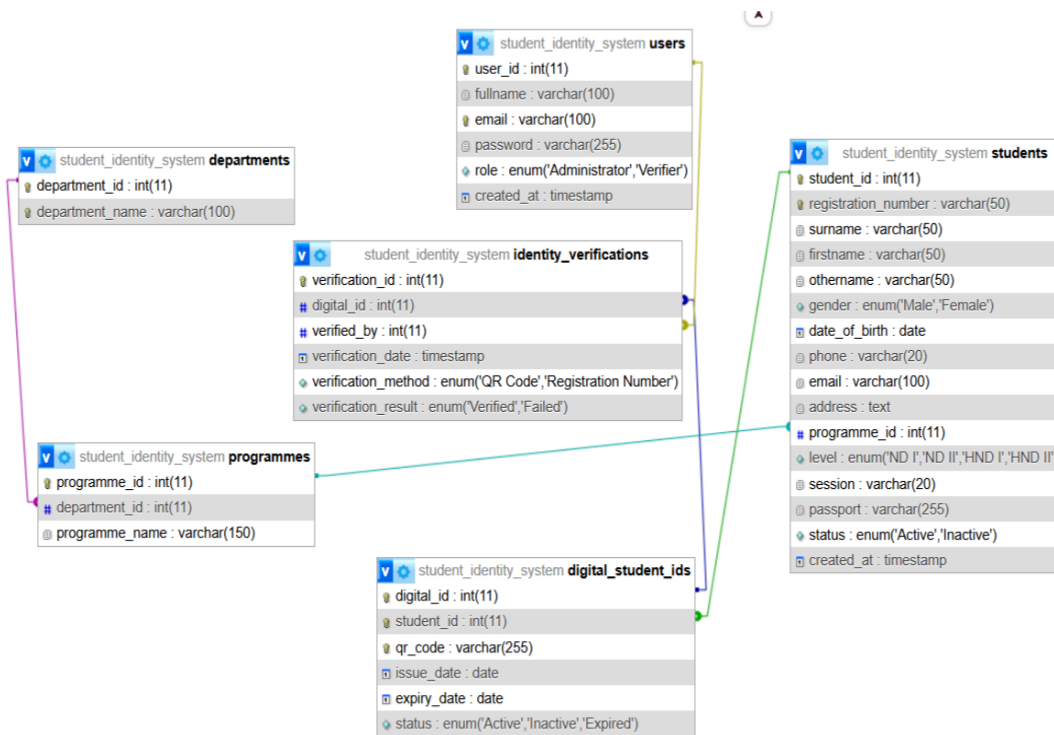
Digital identity information is stored separately through the Digital ID structure. Verification activity is also recorded independently so that verification history can be maintained without altering the underlying student record.

Notifications are associated with user accounts, allowing the system to display notifications according to the logged-in user.

Audit records provide a history of important administrative actions.

The major entities in the implemented database are:

1. Users
2. Students
3. Faculties
4. Departments
5. Programmes
6. Academic Sessions
7. Student IDs
8. Verification Logs
9. Notifications
10. Audit Logs



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Figure 3: Entity Relationship Diagram of DASIS.

The relational design establishes connections among the major entities while reducing unnecessary duplication. Primary keys uniquely identify records, while foreign keys connect related records.

For example, the student record is associated with a user account, a student may be associated with a faculty, department, programme, and academic session, and the Digital ID is linked to the corresponding student.

This arrangement reflects the actual identity-management process of DASIS rather than treating each piece of information as an independent record.

3.4.2 Data Design

Data design describes how information enters the system, how it is processed, where it is stored, and how it is returned to the user.

In DASIS, the principal categories of data are:

1. **Student Personal Data:** This includes the student's name, photograph, date of birth, gender, address, and telephone number.

2. **Academic Data:** This includes the student's matriculation number, faculty, department, programme, level, and academic session.
3. **Account Data:** This includes the user's email, account role, password information, and account status.
4. **Digital Identity Data:** This includes the Digital ID record, card number, QR token, issue date, expiry date, and Digital ID status.
5. **Verification Data:** This includes information about the student being verified, the verification method, the verification result, and the date and time of verification.
6. **Notification Data:** This contains information required to deliver system notifications, including the recipient, title, message, type, read status, and time.
7. **Audit Data:** This contains information about administrative activities performed within the system.

The overall data movement can be represented as:

User Input → Application Processing → Database Storage → Data Retrieval → System Response

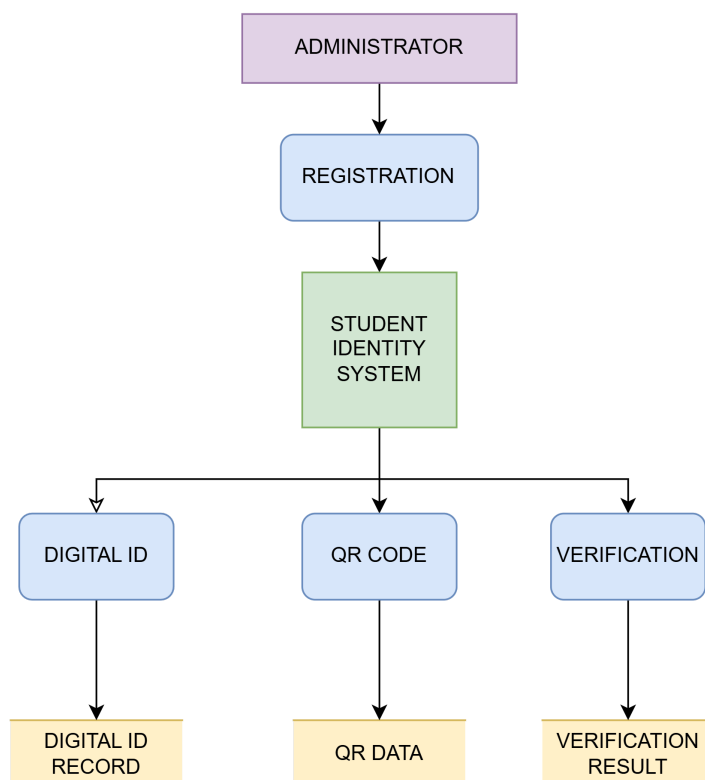


Figure 4: Data flow and organization within DASIS.

The data design ensures that student information can be updated without recreating the entire identity record. This is particularly important for identity management because a student's status, programme, level, or other attributes may change during the period in which the student remains registered.

3.5 Design Considerations

The system design considered several factors that directly affected the architecture and implementation of DASIS.

1. **Security:** Security was considered because student identity information should not be accessible to unauthorized users. Authentication and role-based access control were therefore implemented so that administrative functions are separated from student functions.
2. **Accuracy:** The system validates important information during registration and uses structured database relationships to reduce inconsistent records. Accurate student information is necessary because the verification process depends on matching presented identity information with the correct database record.
3. **Usability:** The interface was designed around clear navigation, visible status indicators, readable information, and straightforward forms. The goal was to reduce the number of steps required for common operations such as registration, viewing a Digital ID, and verifying a QR code.
4. **Data Integrity:** Primary keys and foreign keys are used to maintain relationships among database tables. This ensures that student identity information remains associated with the correct user, academic structure, Digital ID, and verification records.
5. **Efficiency:** QR-code verification provides a faster way to locate a student record compared with manually searching through a large collection of student records. The system also retrieves relevant information electronically rather than requiring repeated manual inspection.
6. **Maintainability:** DASIS separates interface logic, application processing, and database operations. This makes individual components easier to modify without redesigning the entire system.

7. **Scalability:** The database structure allows additional student records and related identity information to be added without changing the basic architecture of the system.

3.5.1 User Interface Design

The user interface was designed according to the roles and tasks performed by the users.

The **student interface** provides access to functions such as:

- Dashboard
- My Profile
- My Digital ID
- QR Code
- Verify My Identity
- Verification History
- ID Status
- Notifications
- Settings

The **administrator interface** provides identity-management and administrative functions including:

- Dashboard
- Students
- Student Verification
- Digital IDs
- QR Management
- Academic Structure Management
- Verification Logs
- Audit Logs
- Notifications
- Settings

The separation of functions ensures that students are presented primarily with services related to their own identity while administrators are provided with management functions. The principal interfaces include the login interface, student registration interface, administrator dashboard, Student Management interface, Digital Student ID, QR verification interface,

verification history, and notification interface. These correspond with the interface categories identified during the original design stage.

3.5.2 Data Design

The data design was implemented to keep personal, academic, identity, and verification information logically connected while avoiding unnecessary repetition.

The student's personal information is stored within the student record, while academic structures are managed through their own related tables. The student's account information is maintained through the user account structure, and Digital ID information is kept separately so that the identity record and the Digital ID status can be managed independently.

This arrangement is particularly important in DASIS because a student's **account status** and **Digital ID status** are not identical concepts. A student account may be active, inactive, rejected, or suspended, while a Digital ID may independently have its own status. Maintaining these concepts separately makes the system more precise when managing identity records.

Verification information is also maintained independently. This allows the system to preserve a history of identity checks even when the student's main information changes.

Notifications are associated with user accounts so that a notification intended for an administrator or student can be retrieved according to the logged-in user.

The data design therefore supports both **current identity management** and **historical record keeping**.

3.5.3 System Design

The final system design is based on a centralized and role-controlled identity-management process.

A student first enters the system through registration and authentication. The registration process collects personal and academic information and creates the necessary identity records.

The administrator can review and manage the resulting student account. The administrator can update account status and manage the student's Digital ID.

After a valid Digital ID is available, the student can access the identity information and QR code associated with the account.

During verification, an authorized user can scan the QR code or provide the registration number. The system retrieves the corresponding student record, checks the relevant identity conditions, and returns the verification result.

Successful and unsuccessful verification activities are recorded for future reference.

Administrative activities are also recorded through the audit mechanism, while system-generated events can be communicated through notifications.

The overall design can therefore be summarized as:

User Authentication → Student Identity Management → Digital ID Management → QR/Registration Verification → Verification Record → Administrative Monitoring

This design directly addresses the central problem identified in Chapter One and reflects the principal concepts discussed in Chapter Two.

CHAPTER FOUR

IMPLEMENTATION, DISCUSSION AND EVALUATION

4.1 System Hardware Requirement

The development and operation of the Digitization of Automated Student Identity System (DASIS) require basic computer hardware capable of running a local web server, database server, web browser, and development tools. The hardware requirements are divided into the development requirements and the requirements for users accessing the system.

The minimum hardware requirements for the development and testing of the system include a desktop computer or laptop with a functional processor, sufficient random-access memory (RAM), adequate storage space, a display monitor, keyboard, and mouse. A smartphone or computer with a camera is also required for testing the QR code scanning functionality.

The hardware requirements are as follows:

Hardware Component	Minimum Requirement	Purpose
Processor	Dual-Core processor or higher	Running the development environment and system
RAM	4 GB or higher	Running XAMPP, browser, database, and development tools
Storage	20 GB of available space	Storing source code, database, and supporting files
Monitor	Standard display	Viewing the system interface
Keyboard	Standard keyboard	Data entry and system operation
Mouse/Touchpad	Standard	Navigation and interaction
Camera	Webcam or smartphone camera	QR code scanning and verification
Network Connection	Local network/Internet where required	Accessing the web application and related resources

The system does not require specialized hardware because it is implemented as a web-based application. The main hardware requirement for the identity verification process is a camera-enabled device capable of scanning the QR code associated with a student's Digital ID.

4.2 System Software Requirement

DASIS was implemented as a web-based application using HTML, CSS, JavaScript, PHP, and MySQL. The technologies selected for the project correspond with the tools described in Chapter Three of this study. HTML provides the structure of web pages, CSS controls their presentation, JavaScript provides client-side behaviour and interaction, PHP handles server-side processing, while MySQL provides relational data storage.

HTML is used to organize interface elements such as forms, tables, navigation menus, buttons, and student information. HTML defines the structure and meaning of web content (MDN Web Docs, 2025).

CSS is used to control the appearance, layout, spacing, typography, and presentation of the application. CSS is specifically designed to describe the presentation of documents written using HTML and related markup languages (MDN Web Docs, 2026).

JavaScript is used to provide interactive behaviour within the system, including dynamic interface operations, communication with the backend API, form interactions, QR-scanning functionality, notification handling, and updating displayed data.

PHP is used as the server-side programming language for the application. It handles authentication, user requests, database operations, student management, digital ID operations, verification, notifications, and administrative operations. The PHP project documentation describes PHP as a general-purpose language especially suited to web development.

MySQL is used as the relational database management system for DASIS. The database stores users, students, academic structures, digital IDs, QR information, verification records, notifications, and audit records. MySQL provides support for relational tables, keys, constraints, and database operations required by the application.

Visual Studio Code is used as the development environment for writing and managing the source code, while XAMPP provides the local Apache and MySQL environment used during development and testing.

A web browser such as Google Chrome is used to access and interact with the system. A camera-enabled device is required when testing the QR code verification process.

The major software requirements are therefore:

Software	Function
Windows Operating System	Development and testing environment
Visual Studio Code	Source code development
XAMPP	Local Apache and MySQL environment
PHP	Server-side application development
MySQL	Relational database management
HTML	Web page structure
CSS	Interface presentation and styling
JavaScript	Client-side interaction and system behaviour
Web Browser	Accessing the DASIS application

4.3 Data Source (Method of Data Collection)

The data required for the development of DASIS were obtained through the observation of existing student identification procedures, examination of the requirements of student identity management, and review of relevant literature relating to digital identity management and QR-code-based verification.

The information collected was used to identify the type of information required for student registration and verification. Such information includes the student's name, registration number, faculty, department, programme, level, academic session, telephone number, date of

birth, gender, address, and photograph. The study also identified information required for the generation and management of Digital IDs. These include the Digital ID record, card number, QR token, issue date, expiry date, and status.

In addition, information required for verification records was identified. This enables the system to maintain records of identity verification activities for monitoring and future reference. The data categories are consistent with the data design presented in Chapter Three. The collected information served as the basis for the database structure, user interface design, identity verification process, and administrative functions implemented in DASIS.

4.4 Implementation Procedure

The implementation of DASIS followed the stages established during the Waterfall development methodology. The system moved from requirements analysis and design to implementation, testing, and preparation for deployment. The Waterfall approach and the associated development stages were described in Chapter Three.

4.4.1 Development Environment

The development was carried out in a local web development environment. Visual Studio Code was used for writing and organizing the HTML, CSS, JavaScript, and PHP source files. XAMPP was used to provide the local Apache web server and MySQL database environment.

The application was organized into a frontend interface and a backend API structure. The frontend communicates with the PHP backend to retrieve and update data in the MySQL database.

4.4.2 Database Implementation

The MySQL database was created to provide persistent storage for the application. The database structure contains related tables for user accounts, students, academic structures, Digital IDs, verification records, notifications, and audit activities.

Primary keys were used to uniquely identify database records, while foreign keys were used to establish relationships between related tables. This approach helps maintain data integrity and minimize unnecessary duplication.

The student record is associated with the user's account through the user identifier. Digital ID information is associated with the corresponding student, while verification activities are associated with the student record and verification process.

4.4.3 User Authentication and Access Control

Authentication was implemented to ensure that only registered users can access protected functions. DASIS provides different access levels based on the user's role.

The principal user roles are:

1. **Student** – accesses personal identity information, Digital ID, QR code, verification functions, verification history, ID status, notifications, and settings.
2. **Administrator** – manages students, Digital IDs, academic structures, verification activities, audit records, notifications, and other administrative functions.

Administrative functions are protected so that ordinary students cannot access restricted management operations.

Passwords are stored using password hashing rather than plain-text storage. PHP's `password_hash()` function creates a password hash using a strong one-way hashing algorithm, with the necessary information included in the generated hash for verification.

4.4.4 Student Registration Module

The student registration module allows student information to be submitted and stored in the database. During registration, the system captures the student's personal and academic information and creates the corresponding account and student record.

The registration process checks required information and prevents duplicate records based on the identity information used by the application.

After successful registration, the student's account initially remains pending/inactive until the required administrative action is completed. A notification is also generated for the administrator when a new student registration is submitted. This allows the administrator to know that a student record requires attention.

4.4.5 Student Management Module

The administrator can access registered student records through the Student Management section. Student information can be viewed and managed from this interface.

The administrator can also change the account status of a student where necessary. Available account states include:

- Active
- Inactive
- Rejected
- Suspended

When an appropriate account action is performed, the corresponding status is saved against the student's account.

4.4.6 Academic Structure Management

DASIS includes administrative management of academic structures. The administrator can manage:

- Faculties
- Departments
- Programmes

These structures are associated with student records so that a student's academic information can be properly represented within the system.

4.4.7 Digital Student ID Module

The Digital Student ID module provides a digital representation of the student's institutional identity.

The Digital ID is associated with the student's record and contains relevant identification information. A unique card number and QR-related information are associated with the Digital ID.

The system also manages Digital ID status. Depending on the administrative action, a Digital ID can be activated, suspended, or revoked.

4.4.8 QR Code Generation and Management

A QR code is associated with the student's Digital ID. The QR code provides a convenient mechanism for retrieving the corresponding student record during verification.

The QR-code concept used by the system follows the general characteristics of QR-code symbology, which specifies encoding, dimensional characteristics, error correction, decoding, and production requirements (International Organization for Standardization, 2024).

The application uses the QR code as an identifier for locating the student's record rather than requiring the verifier to manually search through student records.

4.4.9 Student Verification Module

The verification module provides two principal methods for checking student identity:

1. QR code verification.
2. Student registration number verification.

For QR verification, the authorized user scans the student's QR code. The encoded token is sent to the verification process, where the corresponding student record is retrieved.

The system then checks the identity information and Digital ID status before displaying the verification result.

A successful verification displays the relevant student identity information. An unsuccessful verification occurs when the required student identity record cannot be found or the identity does not satisfy the system's verification conditions.

4.4.10 Verification History

DASIS records verification activities so that identity verification events can be reviewed later. The verification history provides information such as the student involved, verification method, verification result, date, and time. This provides an audit trail of identity verification activities and supports administrative monitoring.

4.4.11 Notification Module

The notification module allows users to receive system-generated notifications.

For the administrator, notifications can be generated when events requiring administrative attention occur, such as a new student registration. The notification interface displays the notification title, message, time, and read/unread state.

The system also supports marking individual notifications as read and marking all available notifications as read.

4.4.12 Audit Log Module

Administrative actions are recorded in the audit log. The audit log provides information about actions performed by authorized administrators, including the action performed, target, date, time, device or related request information, and status.

This provides an additional layer of accountability when administrative changes are made within the system.

4.4.13 Reports and Administrative Records

The administrative section provides access to system records and reports relating to student registration, Digital IDs, verification activities, and other stored information.

The reports component is designed to assist the administrator in retrieving summarized information from the DASIS database.

4.5 Algorithm/Pseudo-codes

4.5.1 Student Registration Algorithm

Algorithm: Student Registration

Input: Student personal and academic information

Output: Student account and student record

BEGIN

1. Receive student registration details.
2. Validate the submitted information.
3. Check whether the email or registration number already exists.
4. IF duplicate information exists:
 - Display registration error.
 - Stop.
5. ELSE:
 - Create user account.
 - Assign student role.
 - Set initial account status to inactive.
 - Create student record.

- Store personal and academic information.
6. Generate/store the student's identity information.
 7. Generate the Digital ID information when applicable.
 8. Create a notification for administrator.
 9. Save all changes.
 10. Display successful registration message.

END

4.5.2 QR Code Verification Algorithm

Algorithm: QR Code Identity Verification

Input: QR token

Output: Verification result

BEGIN

1. Receive QR token.
2. Search the database for a Digital ID associated with the token.
3. IF no matching Digital ID is found:
 - Record failed verification.
 - Display "Verification Failed".
 - Stop.
4. Retrieve the associated student record.
5. Check the Digital ID status.
6. Check whether the identity is valid for verification.
7. IF the identity is not valid:
 - Record failed verification.
 - Display unsuccessful verification result.
 - Stop.
8. ELSE:
 - Retrieve student identity information.
 - Record successful verification.
 - Display verified student information.

END

4.5.3 Student Account Status Algorithm

Algorithm: Update Student Account Status

Input: Student ID and new account status

Output: Updated student account

BEGIN

1. Administrator selects a student.
2. Receive student ID and requested status.
3. Validate student ID and status.
4. Retrieve the student's account record.
5. If the student record does not exist:
 - Display error message.
 - Stop.
6. Update the associated user account status.
7. IF status is Active:
 - Ensure that the student's Digital ID is active.
8. IF status is Suspended:
 - Suspend the student's Digital ID where applicable.
9. Generate appropriate student notification.
10. Record the administrative action in the audit log.
11. Save the changes.
12. Display successful status update.

END

4.5.4 Digital ID Status Algorithm

Algorithm: Digital ID Status Management

BEGIN

1. The administrator selects a student's Digital ID.
2. Select required action:

Activate

Suspend

Revoke

3. Retrieve the student's Digital ID.

4. IF Digital ID does not exist:

Create/retrieve the appropriate identity record.

5. Apply selected status.

6. Update issue/expiry information where applicable.

7. Notify the affected student.

8. Record the administrative action.

9. Save the changes.

10. Display successful operation.

END

4.6 Sample Implementation Input Snapshot

The implementation input screens provide the interfaces through which users supply information to DASIS. The major input interfaces include the login form, student registration form, administrative data forms, and QR verification interface.

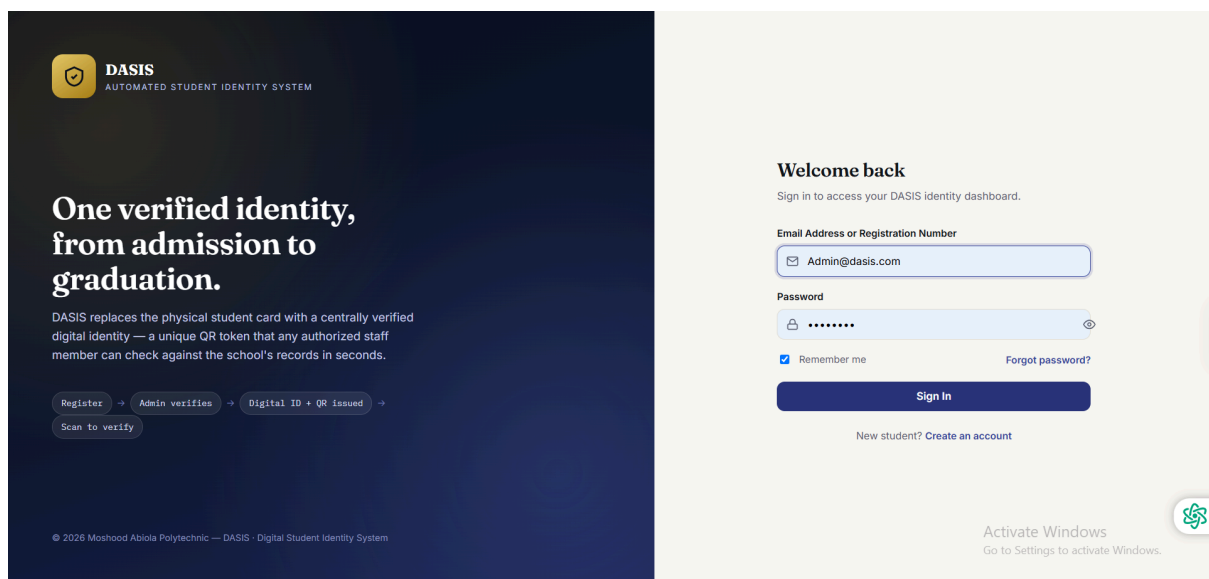
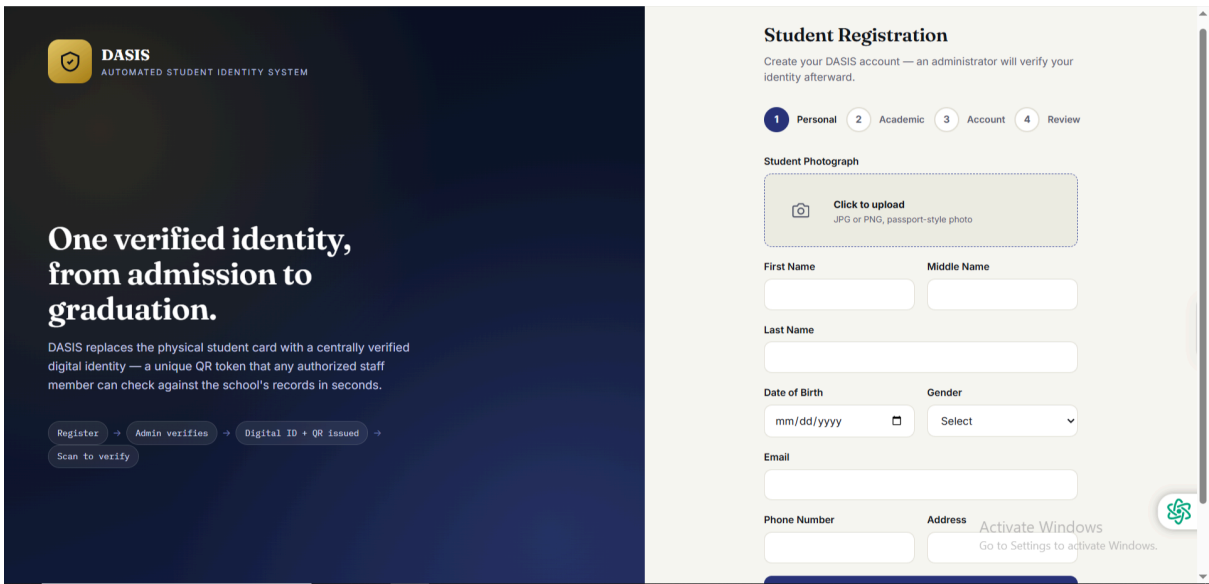


Figure 5: Login interface of the Digitization of Automated Student Identity System.

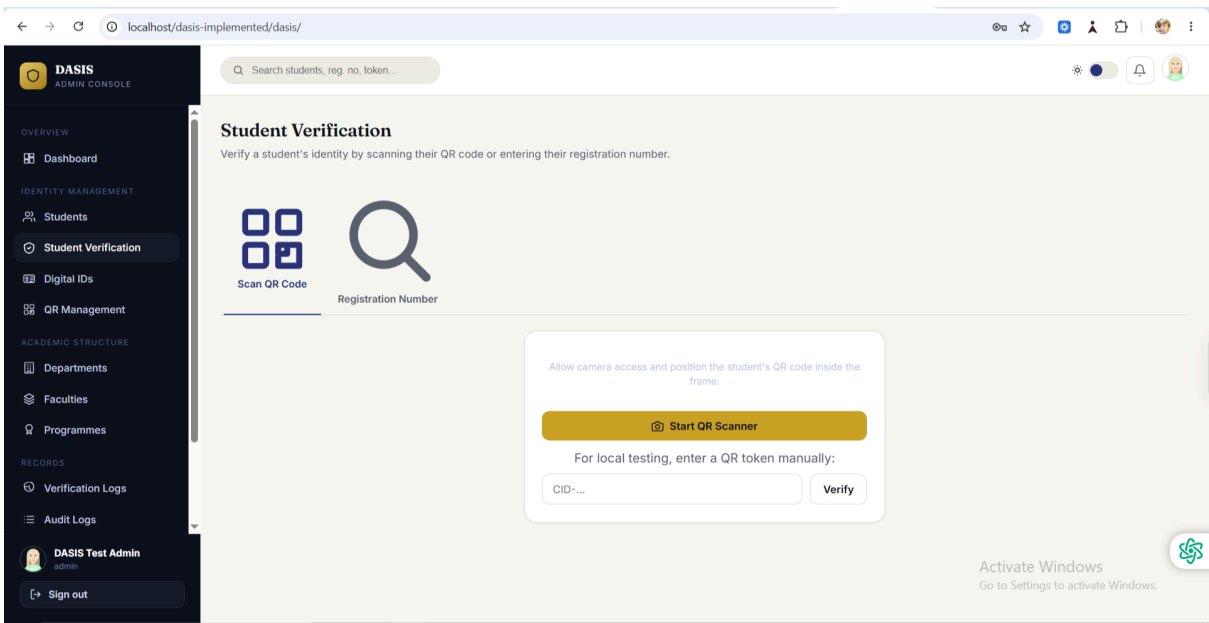
The login interface provides fields for the user to enter the credentials required to access the system. Authentication determines the user's role and grants access to the appropriate system functions.



The screenshot shows the 'Student Registration' page of the DASIS (Automated Student Identity System). The left sidebar features the DASIS logo and a navigation menu with options like 'Register', 'Admin verifies', 'Digital ID + QR issued', and 'Scan to verify'. The main content area is titled 'Student Registration' and includes a progress bar with four steps: 1. Personal, 2. Academic, 3. Account, and 4. Review. Below the progress bar, there is a 'Student Photograph' section with a 'Click to upload' button. The form includes fields for 'First Name', 'Middle Name', 'Last Name', 'Date of Birth' (with a date picker), 'Gender' (a dropdown menu), 'Email', 'Phone Number', and 'Address'. An 'Activate Windows' watermark is visible in the bottom right corner.

Figure 6: Student registration interface for capturing personal and academic information.

The registration interface captures the student's identity and academic information required to create a student record.



The screenshot shows the 'Student Verification' page of the DASIS Admin Console. The left sidebar contains a navigation menu with options like 'Dashboard', 'Students', 'Student Verification', 'Digital IDs', 'QR Management', 'Departments', 'Faculties', 'Programmes', 'Verification Logs', and 'Audit Logs'. The main content area is titled 'Student Verification' and includes a search bar. Below the search bar, there are two options: 'Scan QR Code' and 'Registration Number'. A large 'Start QR Scanner' button is prominently displayed. Below this button, there is a text input field for 'CID-...' and a 'Verify' button. An 'Activate Windows' watermark is visible in the bottom right corner.

Figure 7: QR code verification input interface.

The QR verification interface provides a camera-based mechanism through which an authorized user can scan a student's QR code.

4.7 Sample Implementation Output Snapshot

The output interfaces display information generated from the submitted inputs and database records.

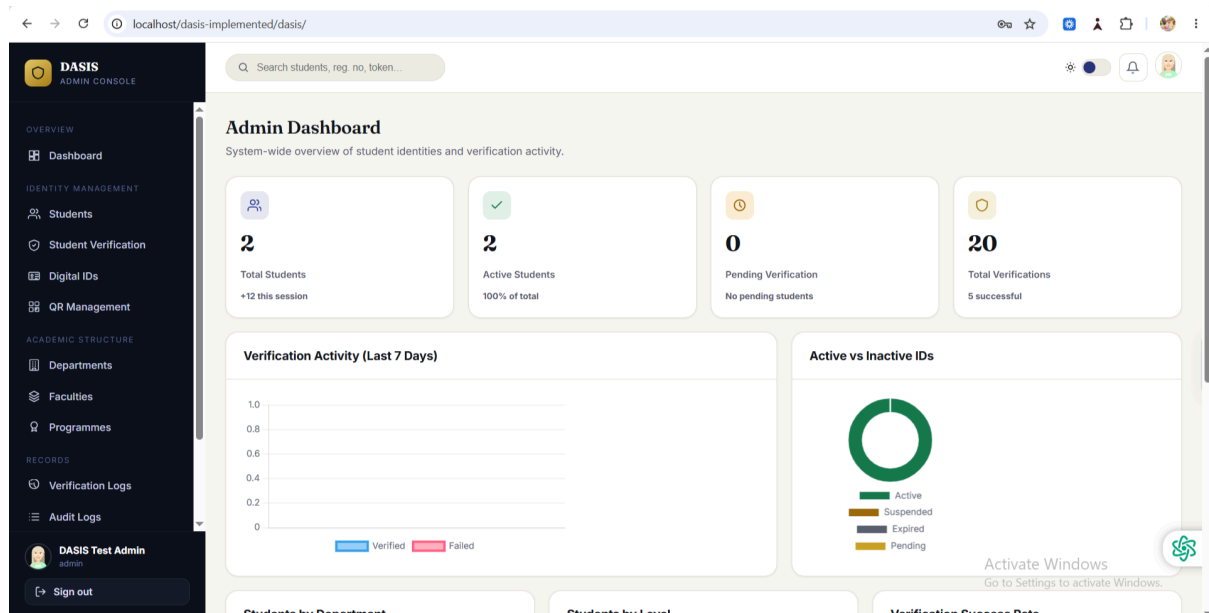


Figure 8: Administrator dashboard of the DASIS.

The administrator dashboard provides a summarized view of important student identity information and administrative activities.

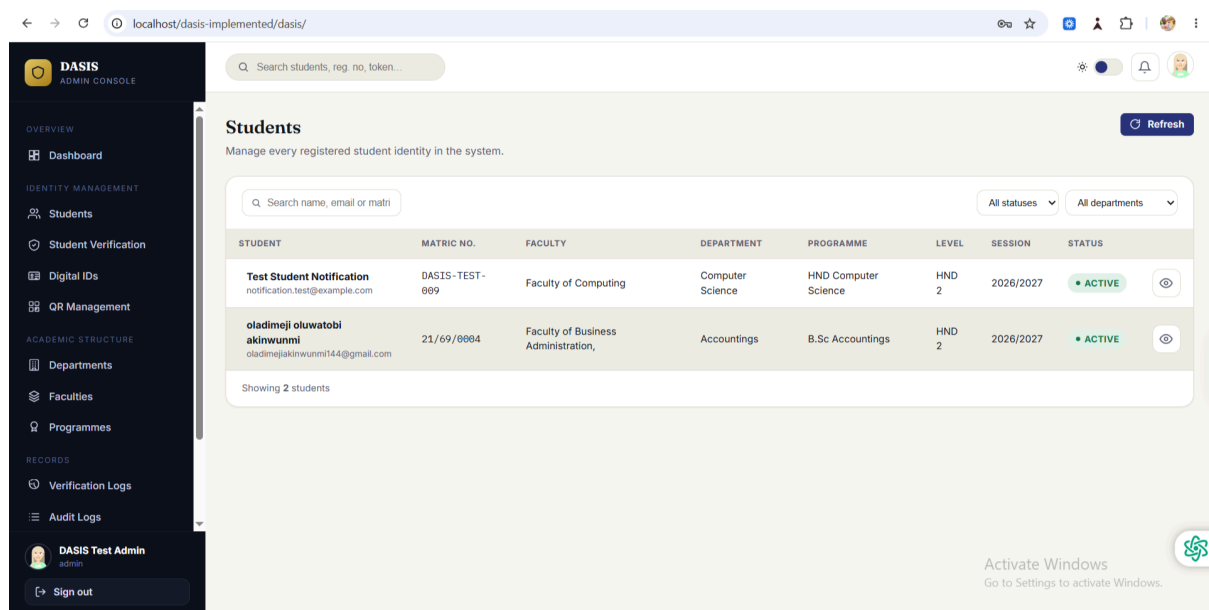


Figure 9: Student management interface.

The Student Management interface displays registered students and provides authorized administrators with controls for viewing and managing student records.

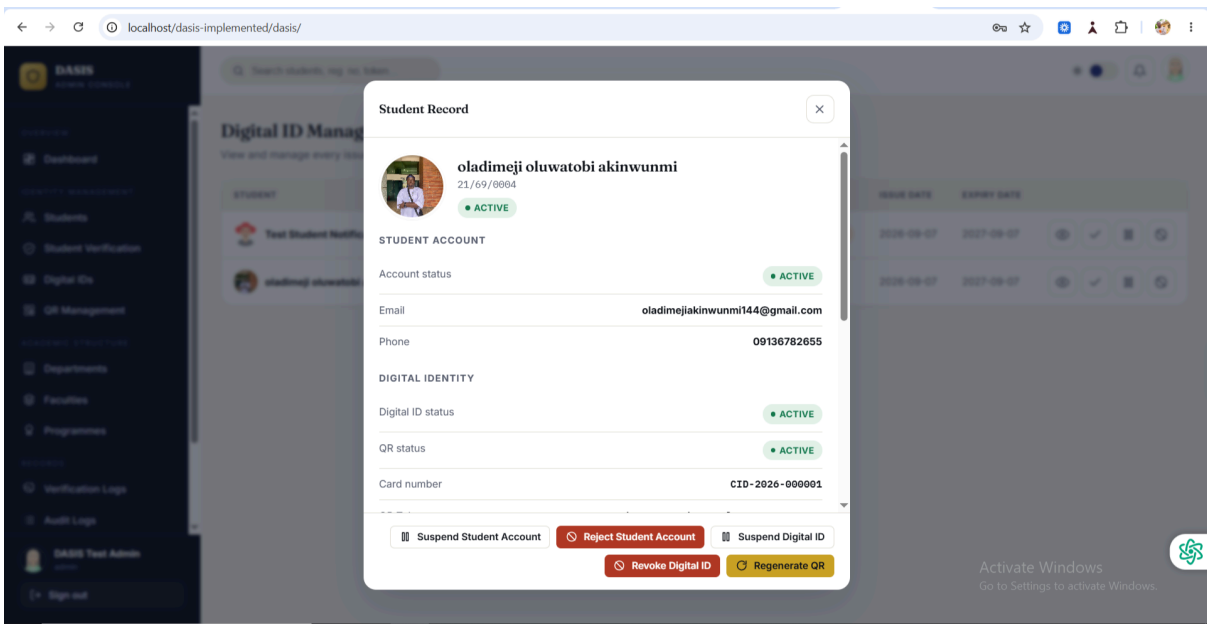


Figure 10: Digital Student ID displaying student identity information and QR code.

The Digital Student ID presents the student's identity information together with the QR code associated with the identity record.

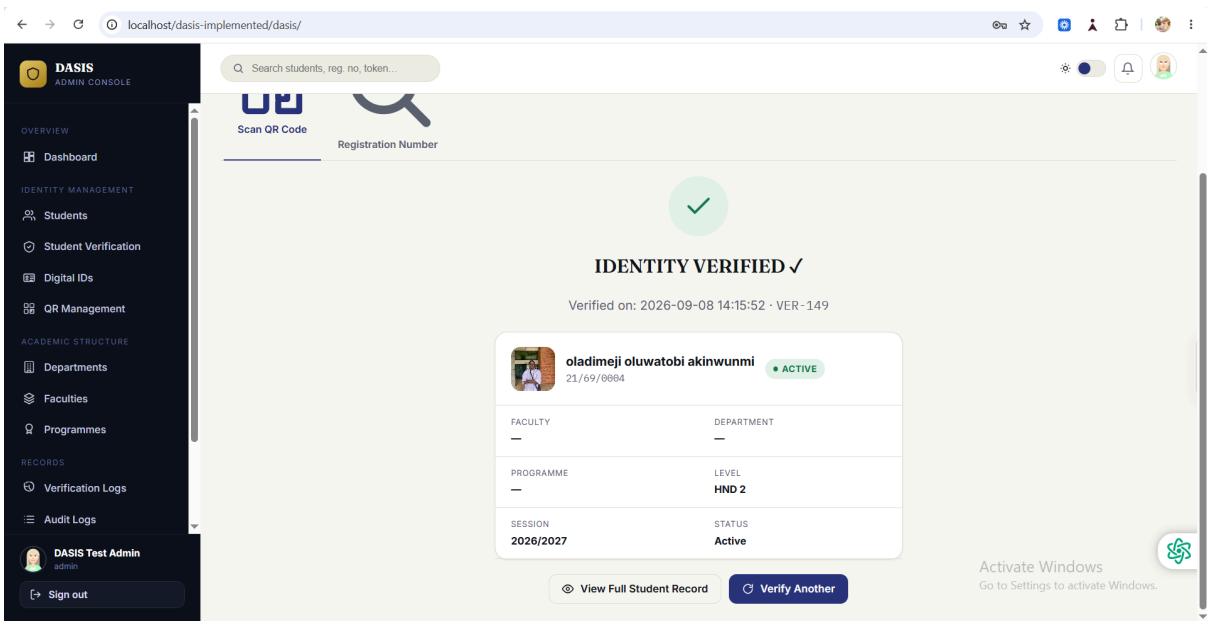


Figure 11: Successful student identity verification result.

The successful verification screen displays the identity information retrieved from the database after the QR code has been successfully matched with a valid student record.

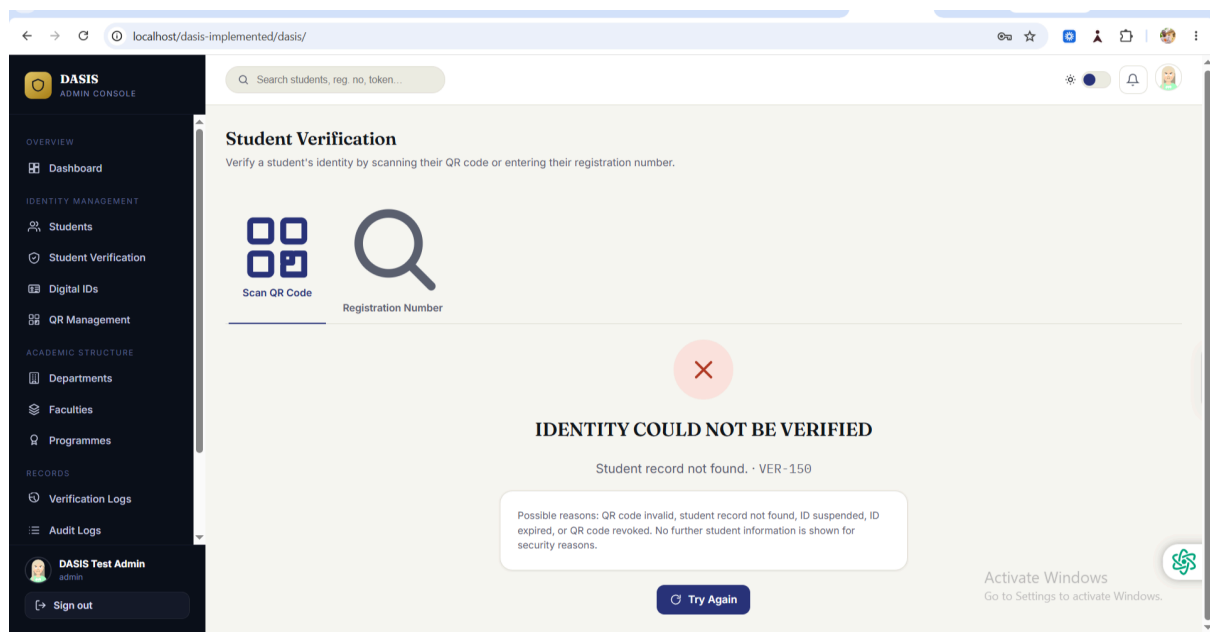


Figure 12: Failed student identity verification result.

The failed verification screen informs the authorized user that the identity could not be successfully confirmed.

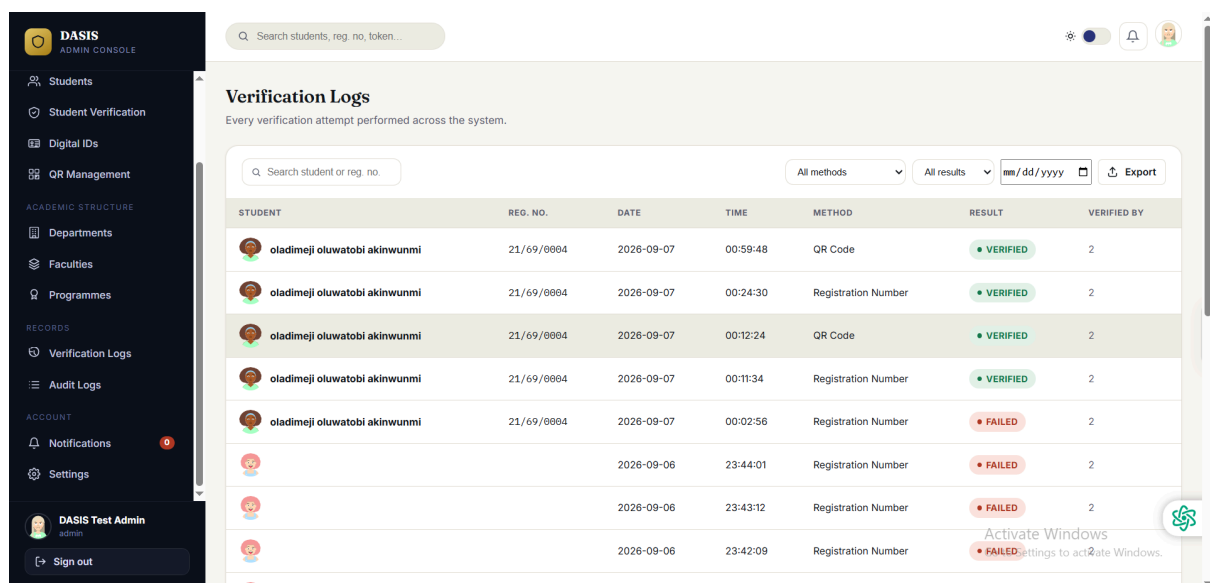


Figure 13: Verification history interface.

The Verification History interface provides a record of previous student identity verification activities.

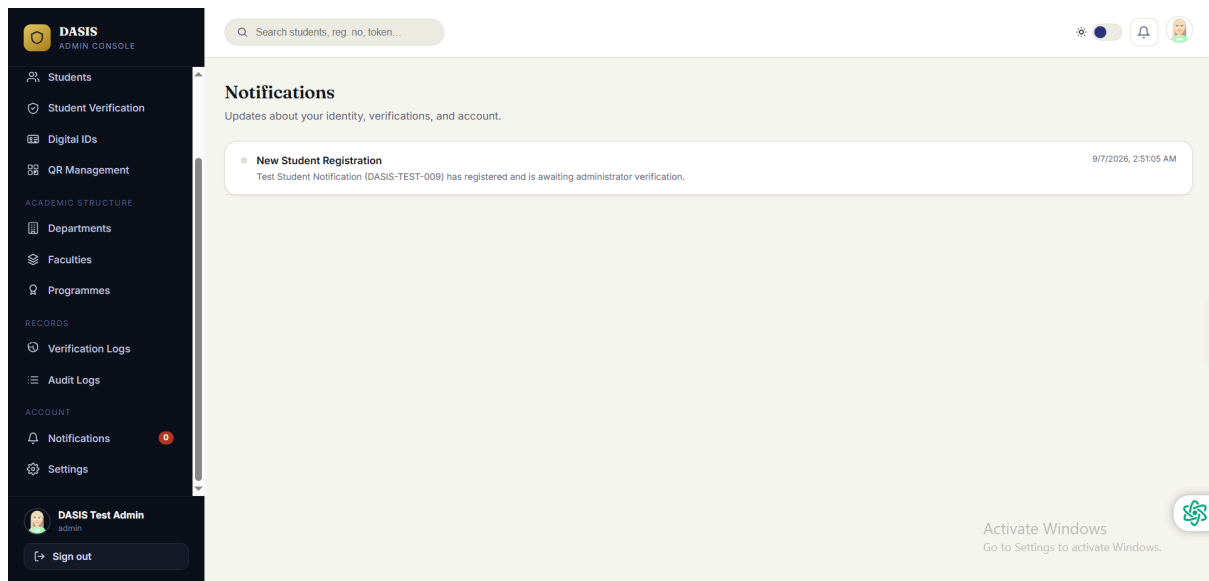


Figure 14: Administrator notification interface.

The notification interface displays system notifications generated for the administrator, including notifications requiring administrative attention.

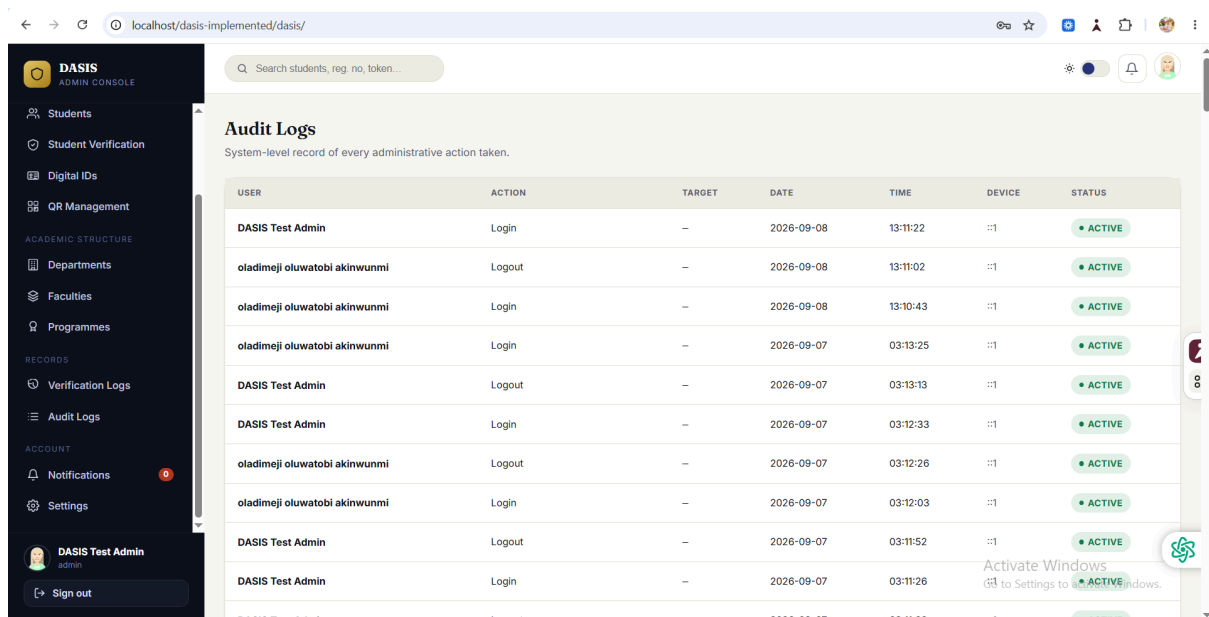


Figure 15: Administrator audit log interface.

The audit log provides a record of administrative activities carried out within the system.

4.8 Evaluation of Results

The developed DASIS was evaluated based on the major functions identified in the objectives of the study. The evaluation focused on whether the system could perform student

registration, identity management, Digital ID generation and management, QR-code verification, verification recording, and administrative management.

The evaluation also considered usability, data integrity, security, and efficiency. These evaluation areas correspond with the objectives of the study, which require the developed system to be tested for functionality, usability, accuracy, and effectiveness in improving student identity management and verification.

Table 4.1: Functional Evaluation of DASIS

S/ N	Function Tested	Expected Result	Evaluation
1	User Login	Valid credentials provide access to the correct user area	Successful
2	Student Registration	Student information is validated and stored	Successful
3	Student Record Retrieval	Stored student information can be retrieved	Successful
4	Student Account Management	Administrator can activate, suspend, or reject an account	Successful
5	Digital ID Management	Digital ID can be managed by an authorized administrator	Successful
6	QR Code Scanning	QR code can be scanned using a camera-enabled device	Successful
7	QR Verification	Valid QR information retrieves the corresponding student identity	Successful
8	Verification History	Verification activities are recorded and displayed	Successful
9	Academic Structure Management	Administrator can manage faculties, departments, and programmes	Successful

10	Notifications	System notifications are retrieved and displayed to the appropriate user	Successful
11	Audit Logs	Administrative actions are recorded for future reference	Successful
12	Access Control	Restricted administrative functions are protected from unauthorized users	Successful

The functional evaluation shows that the major processes implemented in DASIS operate according to the intended system design. In particular, the QR verification process provides a direct mechanism for retrieving a student's identity information rather than requiring the verifier to search manually through records. This is consistent with the design described in Chapter Three, where the process moves from QR scanning to student record retrieval, identity verification, and storage of the verification record.

Table 4.2: Evaluation Against Project Objectives

Objective	Implementation Outcome
Review relevant literature	Relevant concepts relating to student identity, digital identification and QR verification were examined
Design the proposed system	Architecture, database design, data design, and user interfaces were developed
Implement the proposed system	Student registration, identity management, Digital ID, QR verification and administrative functions were implemented
Test and evaluate the system	Functional evaluation was carried out on the principal modules of the developed system

The evaluation indicates that DASIS addresses the principal problems identified in the study. The system centralizes student identity information, provides a Digital ID, associates each identity with QR information, and provides a mechanism for automated identity verification. These functions directly correspond to the original scope of the system.

4.8.1 Analysis of Findings

The implementation demonstrates that student identity management can be organized into a centralized computerized system. Student information can be stored in a structured relational database and retrieved when required.

The Digital ID provides an electronic form of student identification, while the QR code provides a quick mechanism for accessing the associated identity record. This reduces the need for lengthy manual searching when verifying student information.

The administrative functions also improve the ability of authorized personnel to manage student records. Administrative account management, Digital ID management, verification records, notifications, and audit records provide additional control over the identity-management process.

4.8.2 Usability Evaluation

The user interface was designed around clear navigation, readable information, organized forms, and easily identifiable controls. The major interfaces include the login interface, administrator dashboard, student registration interface, Student Management interface, Digital Student ID interface, QR verification interface, and Verification History interface.

The separation between student and administrator interfaces also reduces unnecessary complexity by presenting users with functions relevant to their roles.

4.8.3 Security Evaluation

Security was considered through authentication, role-based access control, controlled administrative functions, password hashing, and audit logging.

The use of password hashing prevents passwords from being stored directly as plain text. Modern password-storage guidance similarly recommends the use of strong password-hashing mechanisms rather than plain-text password storage (OWASP Foundation, 2025).

The use of role-based access also helps restrict administrative operations to authorized users. Audit records provide additional accountability by recording important administrative actions.

4.8.4 Overall Result

The evaluation demonstrates that the developed DASIS successfully provides a centralized platform for student identity management and automated verification. The system combines student registration, structured identity records, Digital IDs, QR code verification, verification history, and administrative controls in a single web-based application.

The resulting system therefore provides an organized alternative to relying exclusively on physical identification cards and manual student-record searches.

CHAPTER FIVE

DISCUSSION, CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The study focused on the design and implementation of a Digitization of Automated Student Identity System (DASIS) for improving student identity management and verification within an educational institution. The study identified challenges associated with traditional student identification methods, including dependence on physical identity cards, manual record searches, loss or damage of physical cards, and the difficulty of quickly confirming student information when a large number of records are involved. These challenges formed the basis for the development of the proposed system.

The developed system provides a centralized platform through which student identity information can be registered, stored, managed, and verified. It provides student registration, student identity management, Digital Student ID management, QR code-based verification, verification history, notifications, and administrative functions.

The use of a Digital Student ID provides a convenient electronic representation of a student's identity. The QR code associated with the Digital ID provides a practical method for retrieving the student's stored identity information during verification. This supports the objective of making identity verification faster and more organized.

The implementation also provides administrators with facilities for managing student accounts, Digital IDs, academic structures, verification records, notifications, and administrative activities. The separation of student and administrator functions helps ensure that users interact with the parts of the system relevant to their roles.

Based on the functional evaluation presented in Chapter Four, the major modules of DASIS operate according to their intended functions. The project therefore demonstrates that a web-based digital identity solution can provide a structured and convenient approach to student identification and verification.

The study concludes that DASIS provides an effective foundation for replacing or reducing dependence on manual student identity verification while still retaining a relatively simple technological structure that can be maintained and extended.

5.2 Recommendations

Based on the findings and implementation of the project, the following recommendations are made. First, educational institutions should consider adopting centralized digital student identity management systems to improve the accessibility and organization of student identity records.

Second, administrators should ensure that student information entered into the system is accurate and regularly updated. Since the effectiveness of the system depends partly on the quality of the stored information, proper data-entry procedures should be maintained.

Third, appropriate security measures should be maintained when deploying the system. Administrative access should remain restricted to authorized users, user credentials should be protected, and audit records should be reviewed periodically.

Fourth, students and identity verification officers should be properly introduced to the use of Digital IDs and QR-code verification so that the system can be used effectively.

Fifth, regular database backups and system maintenance should be carried out to reduce the risk of data loss and ensure continued availability of the system.

Finally, the system should be deployed alongside appropriate institutional policies governing student identity information, access permissions, privacy, and responsible use of identity records.

5.3 Contribution(s) to Knowledge

The project contributes to the area of computerized student identity management by bringing together several identity-management functions within a single system.

The first contribution is the integration of student registration and Digital Student ID management. Instead of treating student registration and identity verification as separate activities, the system links the student's personal and academic information to a centralized identity record.

The second contribution is the use of QR code-based verification as a direct mechanism for retrieving student identity information. The QR code provides a practical link between the Digital Student ID and the student's stored institutional record.

The third contribution is the integration of verification history and administrative auditing. This allows verification activities and administrative actions to be recorded for monitoring and accountability.

The fourth contribution is the provision of role-based functionality for students and administrators. Students can access their identity-related services, while administrators are provided with management tools for controlling student identity records.

The project therefore provides a practical model for implementing a web-based student identity management system that combines digital identification, centralized records, and QR-based verification.

5.4 Suggestions for Further Research

Future researchers and developers can extend DASIS in several directions.

One possible area of future work is the integration of biometric verification technologies such as fingerprint or facial recognition. Such technologies were outside the scope of the present study, which focused on registration numbers, Digital IDs, and QR codes.

Another area is the development of a dedicated mobile application through which students can access and present their Digital IDs directly from mobile devices.

Future research can also investigate offline or low-connectivity verification mechanisms so that identity verification can continue when Internet or server availability is limited. The current study recognizes web/network availability as one of its limitations.

Further work may also investigate stronger institutional integration, allowing the identity system to exchange relevant information with other authorized academic or administrative systems.

Finally, future research may evaluate DASIS using larger numbers of users and longer periods of operation to measure performance, scalability, usability, and security under real institutional conditions.

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